

Living Documents: Designing for User Agency over Automated Text Summarization

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ABSTRACT

Text summarization is an example of a complex algorithm with a single direct input to output user operation. As different types of users e.g. students, journalists, etc. adopt these algorithms, it is imperative that user experience with text summarization AI systems is modified to allow agency. To explore this, we designed an interactive multi-document summarization system, called *Living Documents*, that allows agency over the summarization process. We evaluated this system in a preliminary study with 25 students identified from Prolific. The students perform summarization tasks using the control functions in our system. Results from this study show that these functions contributed to the user's understanding of how summarization works and this understanding slightly impacts user trust in our system positively. We discuss our experience designing for user control of a normally automated, complex process that produces immutable output and the future implications for other applications.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

KEYWORDS

User Agency, Text Summarization System, Interaction Design

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1 INTRODUCTION

Users are encountering an increasing number of complex algorithms in their daily lives, this has impacted a wide range of applications and has resulted in calls for *algorithmic transparency* in a number of domains [4, 6, 10]. Users rarely understand how these complex algorithms function and rarely know how to affect them

to better support their own interests and goals. This combination of factors means that it is important to understand how to give users more agency and control over the complex algorithms that they use.

In this paper, we focus on *text summarization* which is part of a family of algorithms where users can have difficulty knowing how or whether a piece of text was generated. We seek to move text summarization from a 'take it or leave it' black box approach, to a system that provides interaction with and iteration of the text being generated. In this way, we are aiming to move away from a single interaction with static text, towards more dynamic, iterative interactions and – at the very least – enable users to fine-tune the generated summary text.

Additionally, a goal of our system is for users to gain an understanding of and about the underlying algorithms as a result of trying to affect their own goals and interests on the system. That is, we first and foremost enable the user to achieve their goals with the system, and as a side effect of this allowance they will learn how to better understand and utilize the underlying algorithms as tools. Therefore, in our evaluations of our system we evaluate (1) what users learned about the topic that they were asked to write about and (2) what they learned about the system and its underlying algorithms. We want users to be in charge of their learning goals, while making use of a complex, automated system. The system we created to achieve this aim is called *Living Documents*.

Living Documents moves away from text summarization as an arcane set of algorithms, and opens text summarization to its users by giving them the tools and functionality to iteratively affect the results of the summarizer. To do this, we built a series of tools and functions, that sit on top of existing text summarization algorithms, that give users control over and insight into how the system works. Our system provides users the ability to, e.g.: edit the produced text, which the system protects from the summarizer; delete whole sections, which the summarizer will not re-introduce and will remove the topic from future corpus constructions; request new sections from the summarizer (and optionally bring in new sources); and add their own sources for the summarizer to incorporate. The application also provides views into the inner workings of the system in via visualizations and interactions. This is all achieved without modifying the underlying algorithms, instead we build on top of these text summarization algorithms, which means that we are able to switch out the text summarization algorithm with minimal development effort.

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In this paper, we present the *Living Documents* tool and the results of its deployment to Prolific – using an abstractive summarization technique – as a learning tool for general topics. We highlight the fact that users were able to control the summarization process, learn about the topic, develop an understanding of the underlying system, and utilize the system functionalities. While users were positive in regards to the system, a few had misgivings that suggested that users do not have total trust in the system yet.

2 LIVING DOCUMENTS ARCHITECTURE

In this section, we introduce and explain the architectural design of the *Living Documents* system. The over-arching design goal in the creation of this system is to ensure users feel in control of the output generated by the text summarization algorithm. For example, a student interested in learning a topic would be able to use our system and then, tailor the output to their field of study.

2.1 User Interface Interactions

In order to communicate to people, who use our systems, that they have the “power” to modify the generated results, we presented the summarization text in a simple word processor [2]. We made some modifications to the word processor, Vue2-editor¹ to fit our purpose and provided additional control functionality on the user interface, which we discuss below.

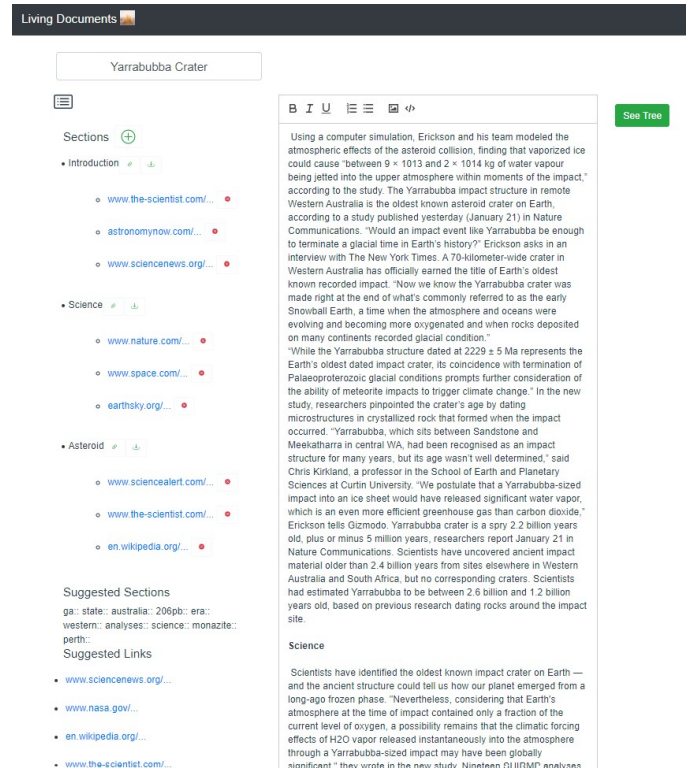
2.1.1 Generating a Document. In the *Living Documents* system, the result returned by the backend is called a document. A document has a title, the summary text, user-defined sections, user-defined or system-generated links leading to the full articles being summarized, system-suggested sections that may fit the topic being summarized, system-suggested links to other similar articles, the topic graph representing the current summary text and a track list of changes made by the user to the summary text.

To initially generate a document, a user provides a topic title they are interested in, with or without links to source articles. This title determines the text our system focuses on in the source articles as well as, driving how we crawl the web for additional articles and similar topic titles – these are displayed respectively in the “Suggested Links” and “Suggested Sections” portions of the interface on the bottom left as shown in Figure 1a.

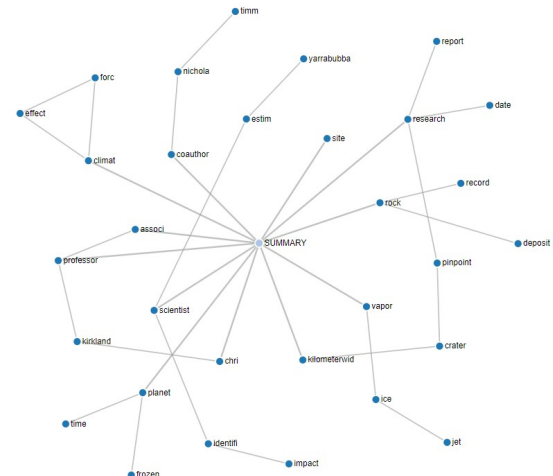
Once the initial summary of the article(s) is provided, users can create new sections to delve deeper into their interests in the topic. A section title can either be selected from the “Suggested Sections” panel or added by the user. These sections will create more specific summaries using the title of the section in conjunction with the title of the document.

2.1.2 Modifying the Document. The user is able to modify the overall structure of the document using agency functions provided by the system. They can add or delete different source article links from the whole document or individual sections. This triggers *Living Documents* to regenerate the document each time so the user can see the impact of their interaction on the document.

At any point in time, the user can make edits and changes to any parts of the document text itself to further reflect their goals and



(a) The Front-end Portion of the *Living Documents* system showing document elements including article links and sections and icons for adding or deleting them.



(b) Example of Concept Graph (shown to users) showing important nodes that contributed to short summary on Yarrabubba Crater

Figure 1: User Interface Interactions

¹(<https://github.com/davidroyer/vue2-editor>) based on the QuillJS editor (<https://quilljs.com/>)

interests. They can edit/delete/add both sections and sentences, all of which impact the underlying model and some of which trigger updates to the document text. Any modifications to section names will trigger the document text to regenerate, while any edits to the text are mostly thought of as corrections to underlying models and affect any future versions of the document. As the user interacts with the system the summary gets more tailored to what they are interested in, additional aspects of the document – aside from just the summary – change with each interaction as well, for example the list of suggested links changes as topics are introduced or pruned.

Any of the edits that the user makes to sentences are protected against being changed or deleted when the user or system triggers an update to the document text. If the edit to a sentence or word is central enough to the constructed topic model, the focus of the document can even change drastically. The user can view this underlying topic model concept graph of the document that the system is maintaining. The graph is meant to give insight into how the system has generated the document and decided which topics are more important or central to the document topic. We anticipate the user will use these insights to make decisions on what types of editions is required to make the document reflect their interests.

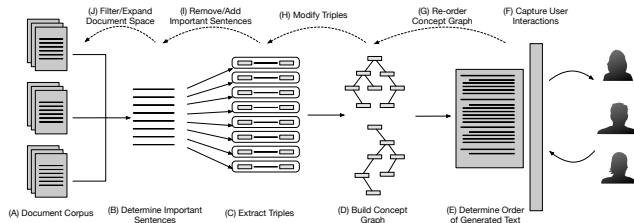


Figure 2: The text processing pipeline of the *Living Documents* system.

2.2 Text Processing Pipeline

The *Living Document* processing pipeline is programmed in Python using a variety of libraries and frameworks, most central of which are Django², the Natural Language Toolkit (NLTK) [7], and the StanfordCore NLP server [9]. In the remainder of the section we describe the components in our pipeline, which is illustrated by Figure 2. Note that this pipeline is iterative, and user interactions are used to modify the underlying structures and document text.

Step A: Construct Document Corpus. When a document is initially created and a title is provided, the system either uses the articles provided by the user or if no article links are provided it searches Google for relevant articles by doing several rounds of query expansion. This initial set of articles (web or PDF documents) are scraped by the system and make up the document corpus. This document corpus is then sent to the summarizer.

Step B: Determine Important Sentences. The manner in which we determine the important sentences is by first, generating a summarization from each of the articles in the corpus using the system’s

²<https://docs.djangoproject.com/en/3.1/topics/migrations/>

current summarizer. The summarizer we use for the preliminary study is an abstractive summarizer called the text to text transfer transformer (T5) model [1, 12]. T5 is pre-trained on a large dataset of news articles. It takes in a text input from a document and returns a text output representing a summary of the article.

At this point, we have a list of sentences representing the summary for each article from either summarizer, however, in our system that does multi-document summarization, we do a post processing step to interleave these sentences, which is described below.

Steps C/D/E: Extract Triples, Build Concept Graph, and Order Sentences. We utilize a combination of triples, graphs, and graph traversals to determine sentence order. This is an important step as multiple articles are being summarized and the document text that the user interacts with needs to flow and make sense.

The first step (*Step C* in Figure 2) is to extract the triples [5, 11] from the collection of important sentences, to extract these triples we use the Stanford CoreNLP Server [8]. A triple is the *Subject, Relation, Object* of a sentence, a sentence may have multiple triples contained in it. These triples represent the smallest units of the summarization results that could be manipulated by the user. Once the triples are extracted and filtered, we use them to build the graph in *Step D* that connects the different sentences via the concepts contained in the triples.

To build the concept graph, we connect the triples via the *Object* node, the *Relation* nodes are not used to find connections between triples. For example, given the sentence ‘*Mary had a little lamb.*’, *Mary* is an *Object*, *had* is a *Relation*, *little lamb* is an *Object* and the resulting graph would be *Mary*→*had*→*little lamb*. If you added the sentence ‘*Bob had a little lamb*’, the resulting graph would be *Mary*→*had*→*little lamb*←*had*←*Bob*. This is because connecting ‘*had*’ would be confusing, but connecting ‘*little lamb*’ helps us to know that both Mary and Bob had a little lamb. We use a variety of techniques to improve these graphs, such as filtering by TF-IDF scores, anaphora resolutions, and stemming. An example of one of these generated graphs is provided in Figure 1b.

To determine sentence order *Step E*, we perform a mix of depth and breadth traversals through the concept graph to keep sentences about the same topic close to each other. Each node in the concept graph is associated with the list of sentences that mention the topic.

Step F/G/H/I/J: Capturing User Interactions and Reflecting them in the System. As described in the *User Interface Interactions* section, the user interface is heavily instrumented. Keeping track of any edits the user makes and the impact of those edits to the underlying concept model is achieved through tying the concept graph (formed from the triples) to the sentences in the interface. In that way, if a user modifies a word that is a central topic to the document, we can make changes and even trigger new document text to be generated or find additional, more relevant documents that they may want to add to the summary. The way this is achieved is by: *Step F* tracking these edits/interactions; *Step G* making any modifications/additions/deletions to the topic model based on these interactions – which may result in pruning entire sections of the graph if sentences or sections or deleted for example; *Step H* modifying/adding/deleting the triples that underlie the graph; and *Steps*

l/j removing sentences or even entire documents from the corpus.

In simpler terms, the *Living Documents* architecture allows a user to “Create a New Document”, set the title of the document, add links related to the topic, after which an initial summary is generated. This summary can help them learn about the topic they are interested in and as they read it they can create new sections to learn more extensively about their desired topic. They can also make text or source link changes to correct or add new information that they find, or make the document read better. Whatever new text the user has entered and the system has generated will be used to modify the document corpus query (e.g. exclude terms that have been deleted and add new terms from new text) and concept graph, so that the suggested portions of the document and future generated summaries more closely reflects the users goals. This process is highly iterative and the user is given feedback at every step to see the impact of their interactions on the generated text.

3 METHOD

To evaluate the *Living Documents*, we recruited 30 participants from Prolific and asked them to use the system on two different topics from different domains. For this evaluation, *Living Documents* uses an abstractive summarizer called T5 [1], the text to text transfer transformer. Each prolific participant that completed the task was paid US \$5.42, we threw out 5 responses that were obviously spam for a total of 25 participants (16 undergraduate and 9 graduate students).

We wanted to specifically investigate whether through using our system: 1) participants learned something about the topic, this tests that text results from the system made sense and that they gained something similar to writing about the topic; 2) participants felt a degree of agency over the system

Topic Selection. The T5 Summarizer has been previously tested with news articles so, we selected topics and news articles related to sports and science. Also, to avoid preconceived opinions about any particular topic, less popular and less controversial topics were used. The sports topics were centered around baseball, and the list of topics ranged from baseball hall of fame, collectable baseball cards and baseball at the Olympics. The science topics were the planet Mars, the Yarrabubba crater and black holes. Each participant performed the evaluation with a single topic from either the sports or sciences.

Tasks and Procedures. The initial task presented to the participants was to answer a 5-level Likert-scale type questions on their knowledge of machine learning, natural language processing and text summarization techniques with responses ranging from not-at-all familiar to somewhat familiar to extremely familiar. They also had to rate their knowledge of their assigned document’s title.

The second task involved performing different actions on the *Living Documents* system. The participants had to create an initial summary using topics and articles they were assigned. Using the suggested sections and suggested articles, the user had the option of delving deeper into their assigned topic. The participants were also instructed to perform interaction techniques such as inserting and deleting sentences from the generated summaries and then to

observe the effects. They could also add or remove articles from any sections of the documents. After each action, the participants were asked to observe changes in the diagram of the concept graphs. Attention check questions were included after the action prompts to ensure that the actions above were completed.

To discover if participants learned anything from reading the generated summary, they were asked to mention the themes and significance of the generated summaries as recommended by Brown et al.[3]. They also rated the qualities of these texts on a 5-level Likert scale ranging from poor to good to excellent. The open-ended questions gave the participants the opportunity to share more about how they influenced and controlled the summarization system, their understanding of the inner workings of the system and functionalities that aided in these understandings.

4 RESULTS

The results from the pre-questionnaire show that the participants were more familiar with machine learning than they were with natural language processing (NLP) and summarization. 23/25 said that they were slightly to extremely familiar with machine learning, while 18/25 expressed some form of familiarity with Natural Language Processing and summarization.

Learning the Topic: The rationale for *Living Documents* was inspired by learners and their need for more self directed, effective learning. Therefore, the most important goal of our system, is for users to be able to learn about a topic that they are interested in. As such, part of evaluating the efficacy of our system was to determine if the users did in fact learn anything about the topic with which they were interacting. One of the topics, used in our evaluation, is whether or not the United States should send professional baseball players to the Olympics in Tokyo. 11 of the 13 participants that were randomly assigned the baseball topic indicated they had no prior knowledge about the topic. After interacting with our system, all the users were able to answer correctly, questions ranging from merely regurgitation to ones that required text interpretation and making an argument about what they learned. Participants attributed their improved knowledge to the quality of the document text. 11/13 participants found the document texts easy to read and understand and this helped them interrogate and learn about the given topic, a main goal of our system. This is an important result, as our method of interleaving text using traversals of concept graphs is a novel way to integrate the summaries of individual documents into a single summary.

User Agency Functions and their effects: The most significant research question in regards to the *Living Documents* system, was to what extent our implementation and design clearly communicated control and agency to its users. At the basic level of interacting with and feeling agency over our system is whether or not users felt as if they could edit the document presented to them, only few participants (4/25) disagreed. 3 of the 25 participants were certain that there interactions with our system had no impact on the summarization results. Even though participants recognized their impacts on the results outputted by the system, only 9/25 were sure that the system reacted adequately to their actions. The number of agency functions made available were fewer than they wanted according to

9 of the 25 participants. Many participants chose neutral positions for some of these enquiries as shown in Figure 3.

In the answers to open-ended debriefing questions, the participants did express some of the control that we were trying to afford them. They understood that the agency functions modified the underlying logic of the system and so, using them contributed significantly to their sense of control. For example, P23 said, “The generated summarizations were dependent on my input, however if the nodes [on the concept graph] are unchanged, there is only a slight change in the summarizations”. Most participants (18/25), including those that mentioned having no or slight knowledge of NLP and summarization techniques, credited their interaction with the *Living Document* tool for helping them know how summarization techniques work. They were able to give high level descriptions of their understanding of the system. According to P3, “[*Living Documents*] sorts words and connectors and finds keywords linked to the title of the document, so it generates a compact text, filled with relevant information”. The system’s logic was transparent to them but this factor did not significantly translate into trust. The levels of trust (15/25) or distrust (7/25) expressed by the participants in the system was motivated by the quality of the document text than the agency functions. That said, some participants’ trust in the system was clearly impacted by their insight into how the system worked. For example, P9’s reason for trusting our text summarization system is, “...because the algorithm looks for keywords, and creates the text using those keywords, and in this way only the summarized text appears”. The control over the summarization process afforded to the participants, had a more indirect and modest impact on participants’ trust over the system.

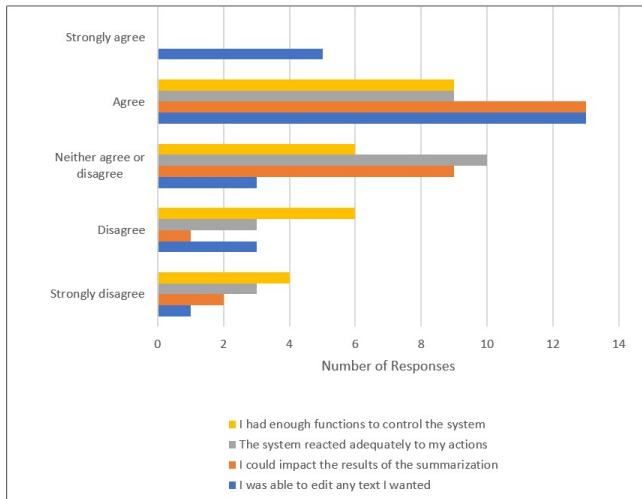


Figure 3: User’s Perceived Control of system with an abstractive summarizer

5 DISCUSSION

The goal of the *Living Documents* system was to open up the black-box of text-summarization and make the process iteratively reactive to user interactions. Our results show promise for the types

of functionality and approaches that we took in our design. If a process as complex as text summarization can be opened up, with no modifications to the base algorithm, i.e., we used an unmodified implementation of the T5 summarizer, there is promise for designing more responsive system for many types of more complex algorithms, without having to make significant modifications to them.

In terms of the questions that we set out to answer, we have a mostly positive set of answers. First, our process shows that, in order to build increased control and feelings of agency over these complex algorithms, it is important to identify the smallest manipulatable units, that can be used to help users understand the results. These units do not necessarily need to be part of the underlying algorithm, they do however, need to be a representation of the results and give levers of control over the underlying algorithm. In our case the algorithm units were the triples extracted from the sentences, we used these to (1) interleave the different document summaries; (2) interpret and reflect the interactions from users; and (3) to visualize the document. We imagine that there are similar units for many complex algorithms. For example, the smallest manipulatable units for facial recognition AI systems may be image pixels.

Participants in our evaluations did seem to gain an understanding of the underlying algorithms by interacting with our system through the functionalities that imparted more control. They either directly or indirectly cited functionalities of our system in their explanations of how the underlying algorithms worked, some deduced the manner in which triples were used to assemble the summaries and many developed a basic understanding of what the system was doing. This means agency functions are a suitable way for making an AI system explainable and transparent.

The control functionalities had a less direct impact on user trust in comparison to the quality of the result. While there is no causation, there is evidence of a correlation between effective agency functions and user trust. The effects and effectiveness of the agency functions we utilize in our system are reflected in the document text result. Our findings point to additional criteria with which we can assess these complex algorithms, that is, instead of focusing only on the result accuracy and utility that an algorithm can come up with on its own without a user, we should move towards thinking about the manner in which the algorithm exposes levers of control to the user.

6 CONCLUSION

This work created a text summarization system to showcase how user agency can be embedded into the functions of a complex algorithm. The building blocks of the system were identified before the system was designed as the triples. Methods for visualizing (the concept graph) and manipulating these triples such as creating of sections, inserting sentences, adding and removing links, were provided to the user. Evaluating the system shows that this development was welcomed by the users and so, this design approach can be employed by future complex systems.

Some participants wanted more of these user controls so, future work can expand and improve functionalities exposed to the user. For instance, in this iteration of *Living Documents* the concept graph was used as merely a visualization of the system, where instead,

the concept graph should be more interactive and support direct manipulation to restructure the document. Some functions that have already been built that could be made available to the users on the UI include the ability to choose the type of summarizer.

REFERENCES

- [1] Yash Agrawal, Atul Thakre, Tejas Tapas, Ayush Kedia, Yash Telkhade, and Vamsundhara Rathod. 2021. *Comparative Analysis of NLP models for Google Meet Transcript Summarization*. Technical Report. EasyChair.
- [2] Robert L Bangert-Drowns. 1993. The word processor as an instructional tool: A meta-analysis of word processing in writing instruction. *Review of Educational research* 63, 1 (1993), 69–93.
- [3] Ann L Brown, Joseph C Campione, and Jeanne D Day. 1981. Learning to learn: On training students to learn from texts. *Educational researcher* 10, 2 (1981), 14–21.
- [4] Motahhare Eslami, Kristen Vaccaro, Min Kyung Lee, Amit Elazari Bar On, Eric Gilbert, and Karrie Karahalios. 2019. User Attitudes towards Algorithmic Opacity and Transparency in Online Reviewing Platforms. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300724>
- [5] Peter Exner and Pierre Nugues. 2012. Entity extraction: From unstructured text to dbpedia rdf triples. In *The Web of Linked Entities Workshop (WoLE 2012)*. CEUR, 58–69.
- [6] René F. Kizilcec. 2016. How Much Information? Effects of Transparency on Trust in an Algorithmic Interface. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 2390–2395. <https://doi.org/10.1145/2858036.2858402>
- [7] Edward Loper and Steven Bird. 2002. NLTK: the natural language toolkit. *arXiv preprint cs/0205028* (2002).
- [8] Christopher Manning, Mihai Surdeanu, John Bauer, Jenny Finkel, Steven Bethard, and David McClosky. 2014. The Stanford CoreNLP natural language processing toolkit. In *Proceedings of 52nd annual meeting of the association for computational linguistics: system demonstrations*. 55–60.
- [9] Christopher D. Manning, Mihai Surdeanu, John Bauer, Jenny Finkel, Steven J. Bethard, and David McClosky. 2014. The Stanford CoreNLP Natural Language Processing Toolkit. In *Association for Computational Linguistics (ACL) System Demonstrations*. 55–60. <http://www.aclweb.org/anthology/P/P14/P14-5010>
- [10] Kenya Freeman Oduor and Eric N Wiebe. 2008. The effects of automated decision algorithm modality and transparency on reported trust and task performance. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, Vol. 52. SAGE Publications Sage CA: Los Angeles, CA, 302–306.
- [11] Delia Rusu, Lorand Dali, Blaz Fortuna, Marko Grobelnik, and Dunja Mladenic. 2007. Triplet extraction from sentences. In *Proceedings of the 10th International Multiconference" Information Society-IS*. 8–12.
- [12] Ekaterina Zolotareva, Tsegaye Misikir Tashu, and Tomás Horváth. 2020. Abstractive Text Summarization using Transfer Learning.. In *ITAT*. 75–80.